

DAY 177/365 • SVM • PART 1 OF 4

Drawing the **widest road** between two classes

Support Vector Machines, explained
simply.

The job: split two groups

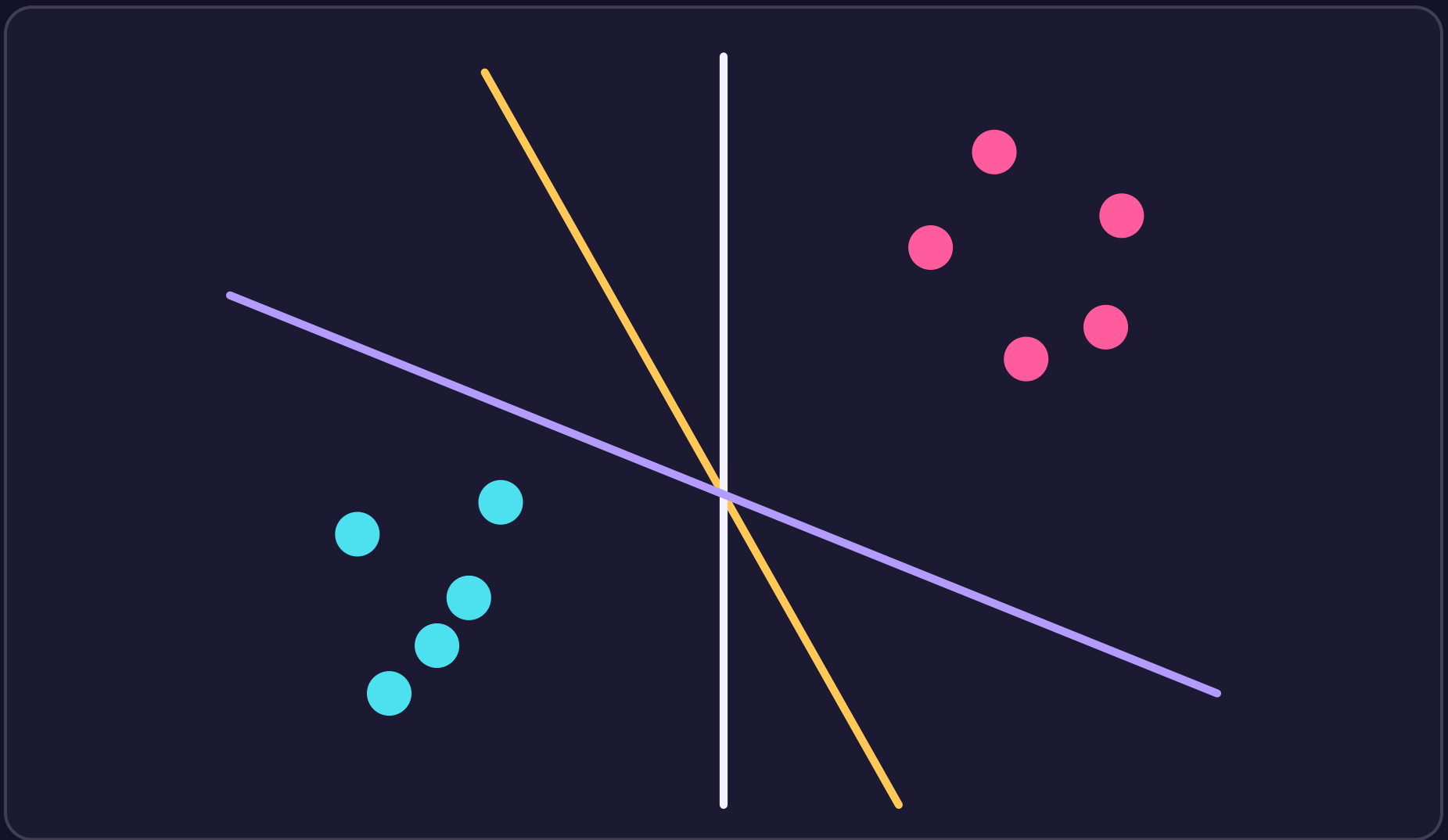
SVM is a supervised model, mostly used for **classification**: spam or not spam, cat or dog, benign or malignant.

Compare with what I've covered

Logistic regression gives a probability. Decision trees ask a chain of questions. **SVM draws one boundary** and tries to draw it in the best possible place.

Today's question: out of all the lines you could draw between two groups, **which is the best one?**

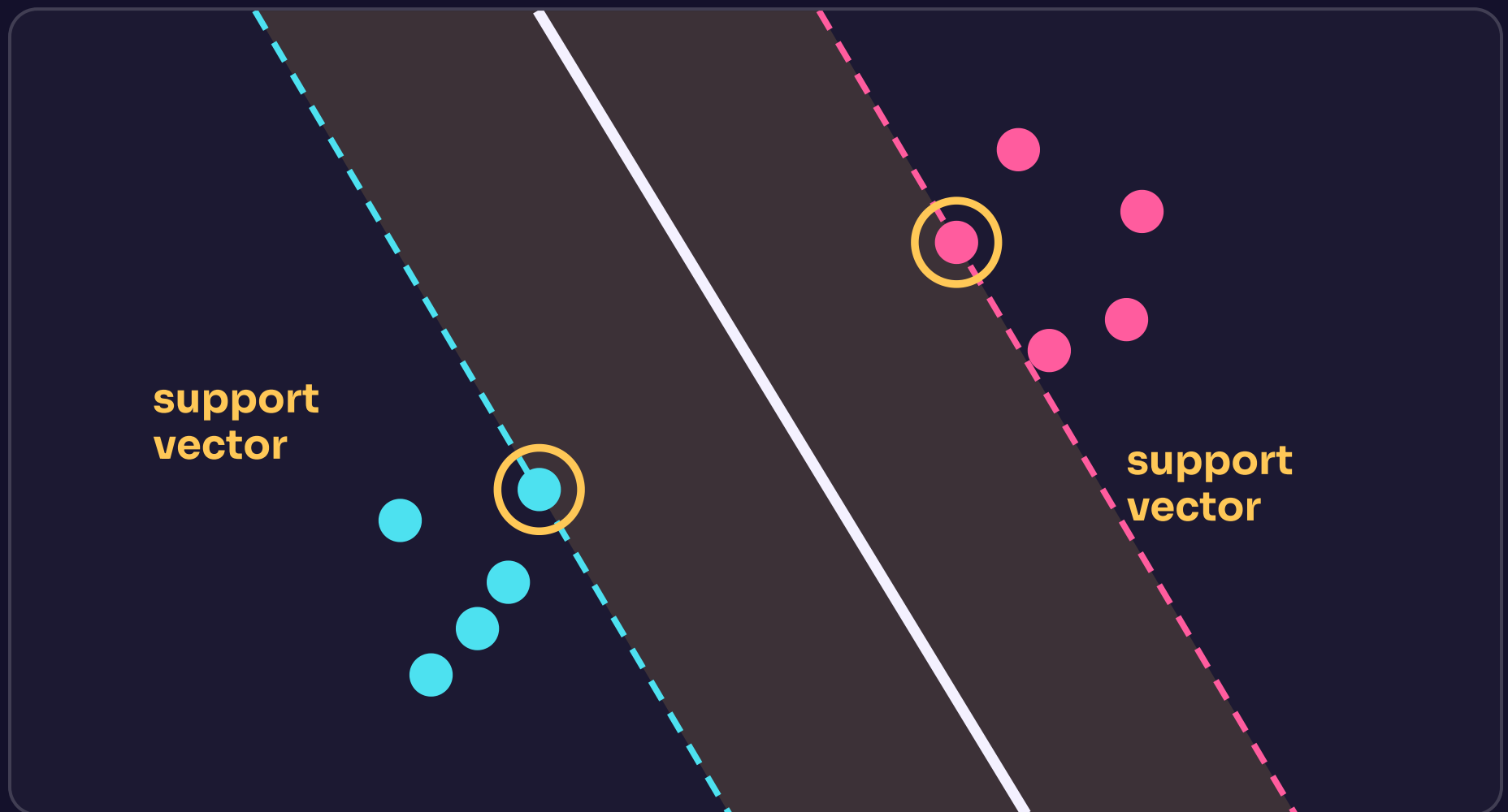
Many lines work. Which one is best?



All three lines separate the training points perfectly. But a new point near the middle could land on the wrong side of some of them.

So we need a rule for picking.

The rule: build the widest road



The solid line is the boundary. The dashed lines mark the **widest empty street** we can fit between the two groups, touching the nearest point on each side.

A wider street means **more room for error**. New points that look a bit different still land on the right side.

Three words to know

Hyperplane

The boundary. A line in 2D, a flat sheet in 3D, and the same idea in higher dimensions. Think of it as the **centre line of the road**.

Margin

The distance from the boundary to the nearest points. It's the **width of the road**. SVM tries to maximize it.

Support vectors

The points closest to the boundary, the ones sitting right on the street's edge. Think of them as the **houses nearest the road**.

Only the nearest points matter

Delete a faraway point

Nothing changes. The road stays exactly where it was.

Nudge a support vector

The road shifts, because those points are what hold it in place.

So out of, say, 1,000 training rows, the final model may depend on only a handful of support vectors.

That's why SVM is described as **memory efficient**: it keeps the points that matter and can forget the rest.

The 1,000-row case is an illustration, not a measured result.

Peek at the maths: one score decides

The boundary is the set of points where a score equals zero:

$$w \cdot x + b = 0$$

Positive score: one class. Negative score: the other. Try $w = (1, 1)$ and $b = -5$:

Point (4, 3)

$4 + 3 - 5 = +2$, so it's on the positive side.

Point (1, 2)

$1 + 2 - 5 = -2$, so it's on the negative side.

Divide a score by the length of w (here $\sqrt{2} \approx 1.41$) and you get the real distance from the boundary: $2 \div 1.41 \approx 1.41$ units.

Illustrative numbers.

TAKEAWAYS

SVM picks the boundary with the **widest margin**.

Hyperplane = the boundary. **Margin** = the road width. **Support vectors** = the points on its edge.

Only the support vectors decide where the boundary goes.

Part 2: what if a point sits on the wrong side of the road? Outliers, the C setting and hinge loss.